From Conflict to Cooperation:
Predicting Gradient Interaction through Representation Alignment

Anonymous Author(s)
Affiliation
Address
email

Abstract

Auxiliary objectives are widely used in deep learning to improve representation quality and generalization, yet their optimization behavior remains poorly understood. In particular, combining a primary task objective with auxiliary semantic supervision often leads to unstable or architecture-dependent gradient interaction. In this work, we show that such a behavior can be understood through a simple geometric principle: the alignment between learned representations and auxiliary targets. We begin with writer identification as a motivating case study, where weak alignment between objectives leads to unstable gradient interaction, sensitivity to loss weighting, and strong dependence on visual backbone architecture. In contrast, compositional image classification exhibits a different regime. Here, representations show strong alignment with auxiliary text targets, leading to consistently positive gradient cosine similarity, negligible conflict, and stable optimization across two different visual architecture backbones. We formalize this phenomenon through a simple alignment statistic measuring the separation between target and non-target similarities. Empirically, we show that this statistic predicts optimization dynamics across tasks and architectures. We further provide a first-order theoretical analysis showing that the expected interaction between objectives is controlled by the alignment.

1 Introduction

Auxiliary supervision is widely used in modern deep learning to improve representation quality, generalization, and robustness. Examples include contrastive learning, prototype alignment, and multimodal supervision. Despite their widespread use, the interaction between objectives remains poorly understood: auxiliary losses can improve performance in some settings, while introducing instability or degradation in others.

A key challenge is understanding when auxiliary objectives cooperate with the primary task and when they interfere with optimization. Prior work has often treated gradient conflict as an inherent challenge in auxiliary objective optimization. In this work, we take a different perspective: we argue that objective interaction is governed by an intrinsic alignment property of the learned representation.

Our investigation is motivated by the writer identification problem prevalent in the document image analysis domain, where an auxiliary text-alignment objective (textual description of the handwriting in terms of visual attributes like slant, cursiveness etc.) exhibits weak interaction with the primary writer-classification objective. In this regime, gradient cosine similarity fluctuates around zero and the behavior depends strongly on the visual backbone. This raises a broader question: is such behavior domain-specific, or does it reflect a more general alignment-controlled mechanism?

To answer this, we provide a predictive characterization of auxiliary representation learning optimization based on alignment, identifying distinct regimes where auxiliary objectives either conflict

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

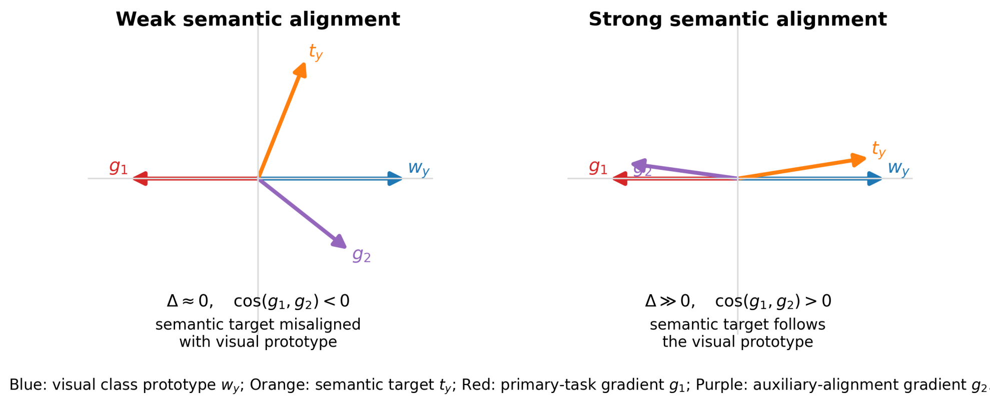


Figure 1: **Alignment controls gradient interaction.** Blue arrows denote the visual class prototype w_y, orange arrows denote the semantic target t_y, red arrows denote the primary-task gradient g_1, and purple arrows denote the auxiliary-alignment gradient g_2. When t_y is weakly aligned with w_y, the two gradients point in conflicting directions. When t_y is strongly aligned with w_y, the gradients become cooperative.

with or reinforce the primary task. In writer identification, weak alignment leads to unstable or architecture-dependent gradient interaction. In compositional image classification, however, representations exhibit strong alignment with auxiliary text targets. These two domains represent contrasting regimes of semantic alignment: writer identification involves loosely structured textual descriptions, while compositional classification uses structured attribute combinations that directly correspond to visual factors. This produces highly positive gradient cosine similarity and negligible conflict, indicating a cooperative optimization regime. **Existing works treat gradient conflict as a property of auxiliary objective optimization and propose heuristic solutions tailored to particular tasks. In contrast, we show that gradient interaction behavior is strongly governed by alignment between representation and auxiliary supervision.** While semantic consistency is intuitively desirable, we show that alignment directly determines the sign and stability of gradient interaction, thereby predicting when auxiliary objectives help or harm optimization.

We formalize this behavior through an alignment statistic that measures the separation between target and non-target similarities. Empirically, we show that this statistic predicts optimization dynamics across different visual representations. In auxiliary representation learning settings, gradient conflict is strongly influenced by alignment between semantic supervision and task structure. When alignment is weak, gradient interactions become unstable and sensitive to hyperparameters. When alignment is strong, gradients become consistently cooperative and auxiliary objectives act as structured regularizers.

Our contributions in this article are as follows:

- We identify semantic alignment as a key factor governing gradient interaction in auxiliary representation learning.
- A simple alignment statistic that predicts the behavior and stability of gradient interaction is introduced.
- Empirical demonstration on alignment-driven regimes across writer identification and compositional image classification is provided to support our proposition.
- We provide a first-order theoretical analysis explaining the transition between conflict and cooperation.

2 Related Work

Multi-objective and multi-task optimization: In multi-objective learning, a primary task loss is typically merged with auxiliary losses through a weighted objective. The central challenge is that gradients from different tasks often point in incompatible directions, leading to interference.

Classical multi-task optimization methods like GradNorm [Chen et al., 2018] normalizes gradient magnitudes to maintain training balance, while MGDA-based approaches frame the problem as formal multi-objective optimization [Sener and Koltun, 2018]. PCGrad [Yu et al., 2020] projects conflicting gradients away from one another to resolve friction. Whereas, CAGrad [Liu et al., 2021] utilizes a conflict-averse update to balance average descent with the improvement of the worst-performing task. Gradient Vaccine [Wang et al., 2021] similarly emphasizes the role of gradient similarity in multi-task optimization.

Contrastive and language-supervised representation learning: Contrastive objectives have become a standard mechanism for aligning representations across modalities. Contrastive Predictive Coding introduced the InfoNCE objective, framing representation learning as density-ratio estimation [Oord et al., 2018]. CLIP showed that large-scale image–text contrastive learning can yield transferable visual features and support zero-shot recognition via natural language prompts [Radford et al., 2021]. Related multi-modal methods such as ALIGN [Jia et al., 2021] further showed the effectiveness of noisy image-text supervision at scale. These methods motivate the use of frozen text embeddings as semantic anchors.

Gradient geometry and representation alignment: Several recent studies highlight the central role of geometry in both optimization and multimodal representation learning. Gradient-based multi-task approaches quantify interactions between tasks through measures such as cosine similarity or gradient conflict [Yu et al., 2020, Liu et al., 2021, Wang et al., 2021]. In parallel, multimodal representation research has revealed structured geometric organization and modality gaps within joint embedding spaces [Liang et al., 2022]. Metric learning methods such as ArcFace [Deng et al., 2019] further emphasize the importance of angular margins and hyperspherical geometry in shaping discriminative embeddings.

Writer identification: Writer identification in the domain of document image analysis is a fine-grained recognition task where classes differ by subtle stroke-level and stylistic variations. DeepWriter, a multi-stream CNN for text-independent writer identification, demonstrated that local handwritten patches can yield discriminative representations of writer identity [Xing and Qiao, 2016]. Building on this, FragNet leveraged fragment-level evidence from word and text-block images to further improve performance [He and Schomaker, 2020]. Together, these approaches highlight that writer identity is encoded in fine-grained visual structure.

Compositional recognition and attribute–object learning: Compositional visual recognition studies how models represent structured concepts such as attribute–object pairs. UT-Zappos50K is a fine-grained shoe dataset introduced for visual comparison and attribute analysis [Yu and Grauman, 2014]. Compositional recognition work such as *From Red Wine to Red Tomato* emphasizes the importance of modeling contextual composition between attributes and objects [Misra et al., 2017]. Subsequent approaches study compositional zero-shot recognition, including attribute-as-operator formulations [Nagarajan and Grauman, 2018] and graph-based open-world compositional learning [Mancini et al., 2022].

Positioning: Our contribution is not in terms of a new architectures, loss-balancing rules, or gradient-surgery algorithms. Instead, we try to explain why auxiliary objectives can be unstable in weak-alignment domains such as writer identification, whereas cooperative in strongly aligned compositional image recognition tasks.

3 Gradient Interaction in Auxiliary Representation Learning

Our analysis provides a first-order characterization of gradient interaction in auxiliary representation learning settings.

3.1 Setup

We consider a model with parameters θ trained with two objectives: $\mathcal{L}(\theta) = \mathcal{L}_1(\theta) + \lambda\mathcal{L}_2(\theta)$, where $\lambda > 0$ controls the contribution of the auxiliary objective.

Let $g_1 = \nabla_\theta \mathcal{L}_1, \quad g_2 = \nabla_\theta \mathcal{L}_2$.

The effective optimization direction is $g = g_1 + \lambda g_2$.

The interaction between objectives is governed by $\langle g_1, g_2 \rangle$.

3.2 First-Order Interaction Analysis

Using a first-order Taylor expansion around θ: $\mathcal{L}_k(\theta+\Delta\theta) \approx \mathcal{L}_k(\theta)+g_k^\top \Delta\theta$, we obtain: $\mathcal{L}(\theta+\Delta\theta) \approx \mathcal{L}(\theta) + (g_1 + \lambda g_2)^\top \Delta\theta$. Thus, local optimization behavior is determined by the geometry of g_1 and g_2. We define: $\cos(g_1, g_2) = \frac{g_1^\top g_2}{\|g_1\|\|g_2\|}$. This yields three regimes: **a) Conflict:** $\cos(g_1, g_2) < 0$; **b)Neutrality:**$\cos(g_1, g_2) \approx 0$; **c) Cooperation:** $\cos(g_1, g_2) > 0$

3.3 Structure of Gradients

For softmax-based objectives, gradients with respect to representation z can be written approximately as: $g_1 \approx \sum_j p_j w_j - w_y$, and $g_2 \approx \sum_k q_k t_k - t_y$,

where w_j are class prototypes, t_k are semantic anchors, and p_j, q_k are softmax probabilities and t_y denotes the target semantic embedding. Each gradient contains attraction toward the target direction and repulsion from competing directions.

3.4 Alignment Statistics and Relation to Gradient Interaction

To quantify the alignment between representations and semantic targets, we introduce an alignment statistic that captures the separation between target and non-target similarities. We define the alignment gap Δ as follows:

Definition (Alignment Gap):- Let $z \in \mathbb{R}^d$ denote the learned representation. Let t_y be the target semantic anchor and $t_{\neg y}$ denote non-target anchors.

$$\Delta = \mathbb{E}[\mathrm{sim}(z, t_y)] - \mathbb{E}[\mathrm{sim}(z, t_{\neg y})].$$

where $\mathrm{sim}(a, b) = \frac{\langle a,b \rangle}{\|a\|\|b\|}$ denotes cosine similarity. Since cosine similarity is bounded in $[-1, 1]$, the alignment gap Δ is also bounded, making it a stable and comparable measure across tasks and architectures. The proposed alignment gap is intended as an interpretable representation-level statistic rather than a universal replacement for existing gradient-conflict or task-affinity measures.

Let us now formalize the role of semantic alignment in controlling the interaction between objectives.

This Δ measures the separation between target and non-target semantic similarities. A small Δ indicates strong overlap (weak alignment), while a large Δ indicates clear separation (strong alignment). We now relate this alignment statistic to gradient interaction.

Approximate characterization of gradient interaction (Alignment Controls Gradient Interaction). Consider two softmax-based objectives sharing a common representation z, with gradients of the form:

$$g_1 \approx \sum_j p_j w_j - w_y, \quad g_2 \approx \sum_k q_k t_k - t_y,$$

where w_j are class prototypes, t_k are semantic anchors, and p_j, q_k are softmax probabilities.

Under mild assumptions that: (i) softmax distributions are moderately peaked around their targets (i.e., most probability mass is concentrated on the correct class), (ii) cross-class correlations are bounded (i.e., non-target similarities do not dominate the target term), the expected gradient interaction can be approximated as:

$$\mathbb{E}[\langle g_1, g_2 \rangle] \approx \alpha\, \Delta - \beta,$$

where $\alpha = (1 - p_y)(1 - q_y) \geq 0$ captures the effective target-alignment contribution, and β aggregates non-target and cross-class interactions, which can have mixed signs but collectively act as interfering terms that reduce the contribution of target alignment. This decomposition suggests the following regimes:(a) **Strong alignment ($\Delta \gg 0$):** The target-alignment term dominates, leading to $\mathbb{E}[\langle g_1, g_2 \rangle] > 0$. Gradients are cooperative and objectives reinforce each other.(b)**Weak alignment ($\Delta \approx 0$):** Non-target and cross terms become significant, and $\mathbb{E}[\langle g_1, g_2 \rangle]$ may be small or negative. Gradient interaction becomes unstable or sensitive to model details.

Connection to derivation. As shown in Appendix A, the gradient interaction decomposes into a target-alignment term $\langle w_y, t_y \rangle$ and multiple competing terms involving non-target prototypes and anchors. The alignment gap Δ serves as an empirical proxy for the dominance of the target-alignment term over these competing contributions.

3.5 Interaction Regimes

Weak Alignment Regime. When $\Delta \approx 0$, target and non-target similarities overlap strongly. Gradient interaction is unstable, often near zero or negative, and optimization becomes sensitive to λ and backbone architecture.

Strong Alignment Regime. When $\Delta \gg 0$, target and non-target similarities are well separated. Gradients consistently align and the auxiliary objective acts as a structured regularizer.

3.6 Critical Interaction Scale

The combined gradient norm satisfies: $\|g_1 + \lambda g_2\|^2 = \|g_1\|^2 + 2\lambda g_1^\top g_2 + \lambda^2 \|g_2\|^2$.

The critical interaction scale is: $\lambda^\star = -\frac{g_1^\top g_2}{\|g_2\|^2}$.

If $g_1^\top g_2 < 0$, cancellation can occur. If $g_1^\top g_2 > 0$, objectives reinforce each other.

3.7 Representation Geometry

Let $z \in \mathbb{R}^d$ be the learned representation. We define:

$$z_V = P_V z, \quad z_T = P_T z, E_V = \|z_V\|^2, \quad E_T = \|z_T\|^2. \tag{1}$$

where P_V and P_T denote projection operators onto the visual and semantic subspaces, respectively. Empirically, we examine whether these quantities remain stable across λ. This allows us to distinguish changes in optimization dynamics from changes in representation geometry.

3.8 Task-Dependent Alignment Regimes

The alignment gap Δ depends not only on the representation but also on the compatibility between the semantic space and the discriminative structure of the task.

Weak Alignment Regime (Writer Identification). In writer identification, the primary task relies on fine-grained visual features such as stroke dynamics, character formation, and writing style. However, the auxiliary semantic space (text embeddings) encodes coarse attributes that are not uniquely discriminative at the writer level. As a result, multiple writers are mapped to similar semantic representations, leading to: $\Delta \approx 0$.

Under this regime, target and non-target similarities overlap significantly, and the gradient interaction becomes: $\mathbb{E}[\langle g_1, g_2 \rangle] \approx 0 \quad$ or fluctuates.

This explains the empirically observed instability and strong dependence on backbone architecture.

Strong Alignment Regime (Compositional Learning). In compositional image classification, the semantic space provides a structured and discriminative decomposition of the visual space. Text embeddings correspond to meaningful attribute-object compositions, which align closely with visual representations.

This leads to: $\Delta \gg 0$, and therefore: $\mathbb{E}[\langle g_1, g_2 \rangle] > 0$.

In this regime, gradients are consistently cooperative, and the auxiliary objective acts as a structured regularizer.

4 Empirical Study: Writer Identification

4.1 Setup

Dataset and Subsampling. We conduct experiments on the IAM Handwriting Dataset, which contains samples from 657 writers. For computational efficiency, we select a pool of 350 writers and randomly sample 300 for training in each run. Experiments are repeated across three seeds, and results are averaged.

To guide learning, we introduce eight anchor-based labels capturing handwriting traits: slant, stroke thickness, character spacing, word spacing, baseline alignment, curvature, connectivity, and ascender/descender patterns. These provide auxiliary supervision, encouraging interpretable and discriminative representations while reducing conflicts during optimization.

All experiments use fixed seeds (28, 12, 2000), controlling initialization and training stochasticity. The data split remains constant, ensuring that variability reflects optimization effects rather than sampling differences.

This setup ensures statistical reliability and tests our hypothesis on instability. Specifically, we expect higher variance across seeds in weak-alignment regimes ($\Delta \approx 0$), while ArcFace-only and adaptive λ methods should exhibit greater stability. Standard deviation across runs thus serves as an indicator of optimization stability.

Models: We evaluate three visual backbones: DeepWriter, FragNet, and a Transformer-based writer identification model.

Objective: We optimize

$$\mathcal{L} = \mathcal{L}_{\mathrm{arc}} + \lambda \mathcal{L}_{\mathrm{nce}},$$

where $\mathcal{L}_{\mathrm{arc}}$ is the ArcFace classification loss and $\mathcal{L}_{\mathrm{nce}}$ is an InfoNCE-based alignment loss using frozen text embeddings as semantic anchors.

Metrics: For each sample, we compute gradients g_{arc} and g_{nce} and measure their cosine similarity. We also compute the alignment gap Δ from target and non-target semantic similarities.

4.2 Weak Alignment Leads to Architecture-Dependent Interaction

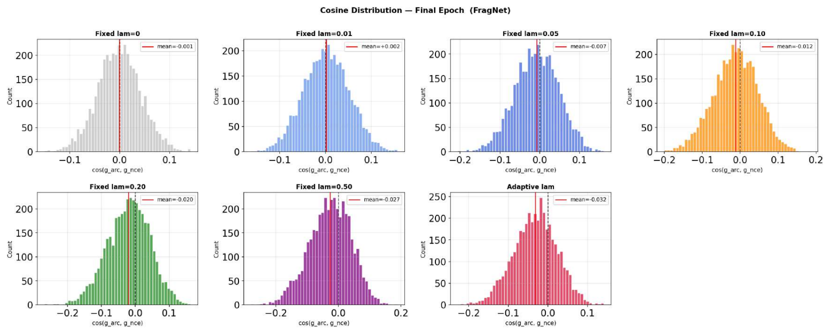
We observe three distinct regimes depending on the visual backbone in figure 2. FragNet exhibits a left-skewed distribution with significant negative mass, indicating frequent gradient conflict. The Transformer remains centered near zero, suggesting weak or neutral interaction. DeepWriter shows a mild right shift, indicating partial cooperation. This variation across architectures indicates that, under weak semantic alignment, gradient interaction is not stable but depends largely on the learned visual representation, rather than being an intrinsic property of the task.

4.3 Weak Target–Non-Target Separation

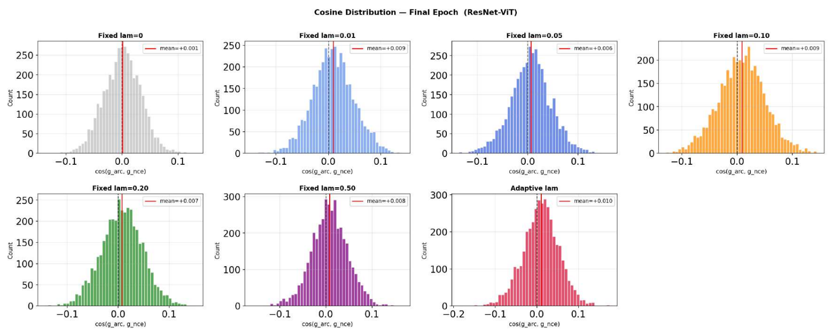
This weak alignment arises because the auxiliary semantic space fails to provide a discriminative partition at the writer level, see figure 3 for an illustration. While text descriptions encode coarse attributes (e.g., slant, spacing), writer identification depends on fine-grained stroke-level variations. As a result, multiple writers share similar semantic anchors, leading to substantial overlap between target and non-target similarities (as shown in figure 3) and a small alignment gap $\Delta \approx 0$. Considering the mentioned phenomenon we can say that writer identification operates in a weak-alignment regime, where target and non-target similarities overlap significantly, leading to unstable and architecture-dependent gradient interaction.

5 Cross-Domain Study: Compositional Image Classification

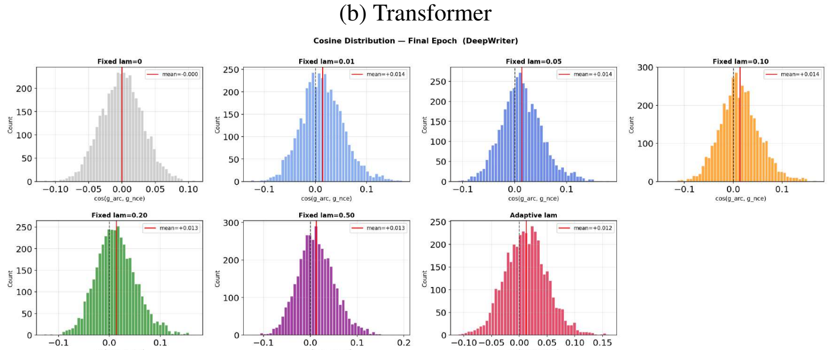
We repeat the same auxiliary objective formulation in compositional image classification using CLIP-based text supervision. We evaluate ViT-B/16, and EfficientNet-B0. In this section, due to space constraint we only report some of the observed experimental results. Details about the experimental setup, dataset and more analysis on results can be found in appendix E.



(a) FragNet



(b) Transformer



(c) DeepWriter

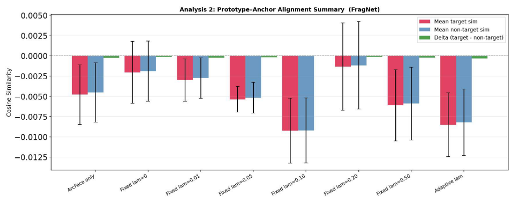
Figure 2: Gradient cosine distributions across architectures in writer identification. FragNet shows conflict, Transformer is near-neutral, and DeepWriter exhibits partial alignment. Please use 300% zoom for better visibility.

5.1 Strong Alignment Emerges

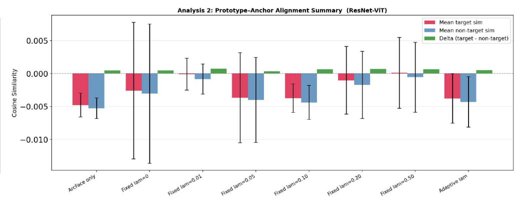
In contrast to writer identification, target and non-target similarities are clearly separated in figure 4, with $\Delta \approx 0.17$, indicating well separation between target and non-target similarities. This indicates that the auxiliary semantic space is well aligned with the visual decision structure of compositional classes.

5.2 Gradient Interaction Becomes Cooperative

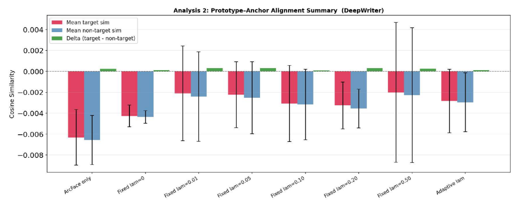
As shown in Fig. 5, gradient cosine similarity becomes strongly positive early in training and remains stable across epochs. The final distribution in figure 6 shows that nearly all samples lie in the cooperative regime, with negligible negative gradient fraction.



(a) FragNet



(b) Transformer



(c) DeepWriter

Figure 3: Target vs non-target similarity summaries in writer identification. Substantial overlap and small target–non-target separation indicate weak semantic alignment.Please use 300% zoom for better visibility.

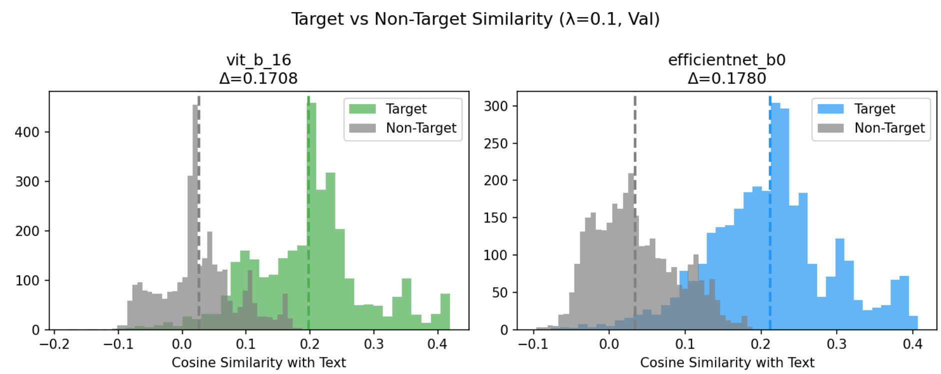


Figure 4: Target vs non-target similarity distributions in compositional image classification. Clear separation indicates strong alignment ($\Delta \gg 0$).

Discussion: The experimental results support a unified explanation that the auxiliary objectives help when they are geometrically compatible with the primary task. In the weak-alignment regime, the auxiliary objective does not reliably reinforce the task objective, and small architectural differences can lead to different gradient regimes. In the strong-alignment regime, the auxiliary objective consistently aligns with the primary loss and behaves as a structured regularizer. This perspective reframes λ not as a simple control over representation structure, but as a scale applied to an interaction whose sign and stability are determined by alignment. Thus, it can be inferred that tuning λ is meaningful only after understanding whether the auxiliary signal is cooperative, neutral, or conflicting.

6 Conclusion

We studied gradient interaction in auxiliary representation learning through the geometry of representation alignment. Starting from writer identification, we observed a weak-alignment regime where gradient behavior is unstable and architecture-dependent. In compositional image classification, we observed a strong-alignment regime where gradients are consistently cooperative across architectures. These results suggest that alignment between learned representations and auxiliary supervision plays a central role in determining whether auxiliary objectives stabilize or interfere with optimization.

We formalized this behavior through an alignment statistic measuring the separation between target and non-target similarities, and provided a first-order theoretical characterization relating alignment

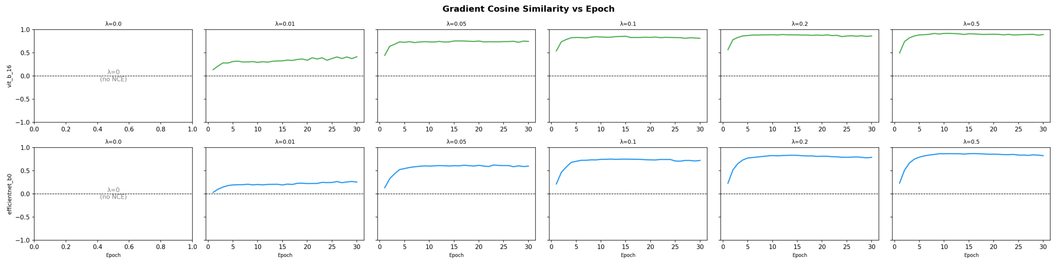


Figure 5: Gradient cosine similarity over training in compositional image classification. Values rapidly become positive and stable.Please use 300% zoom for better visibility.

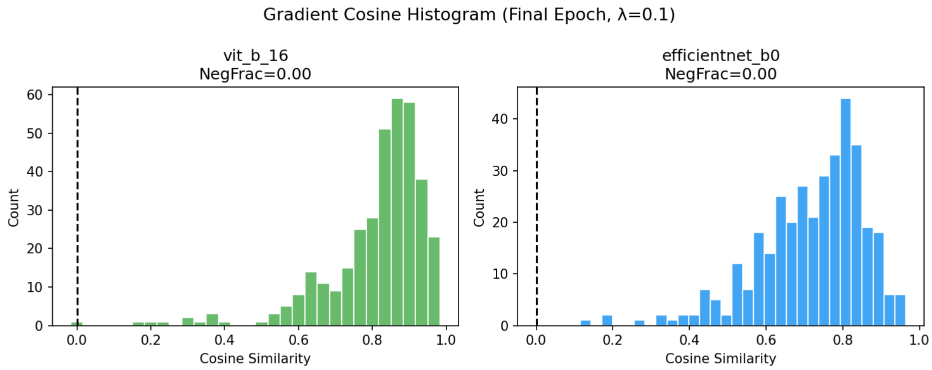


Figure 6: Final gradient cosine distribution in compositional image classification, showing strong cooperative interaction.

to expected gradient interaction. Empirically, we demonstrated that this alignment-driven view consistently explains optimization behavior across different architectures and auxiliary supervision regimes.

7 Limitations and Future Work

Our analysis provides a first-order characterization of gradient interaction in auxiliary representation learning settings and relies on several mild assumptions.

The proposed alignment statistic is formulated using pairwise similarity structure between learned representations and discrete semantic anchors. As a result, the current formulation is most naturally suited to auxiliary supervision settings involving categorical or attribute-based semantic structure. Extending the framework to continuous targets (regression problems), dense prediction tasks (such as segmentation or depth estimation), or broader multi-task settings may require more general notions of representation alignment and gradient interaction.

Secondly, the current theoretical analysis focuses on expected first-order gradient interaction and does not explicitly model higher-order optimization effects or training trajectory dynamics.

Finally, while the experiments demonstrate consistent alignment-controlled behavior across writer identification and compositional image classification, broader validation across additional architectures, datasets, and forms of auxiliary supervision would further clarify the generality and limits of the proposed framework. A systematic comparison with broader task-affinity metrics across diverse optimization settings remains an interesting direction for future work.

8 Acknowledgment

We used AI-based tools to assist in drafting the manuscript and improving the clarity of the mathematical presentation. All derivations, results, and claims were independently verified by the authors.

References

Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 794–803. PMLR, 2018.

Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4690–4699, 2019.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Sheng He and Lambert Schomaker. Fragnet: Writer identification using deep fragment networks. *IEEE Transactions on Information Forensics and Security*, 15:3013–3022, 2020.

Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning with noisy text supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4904–4916. PMLR, 2021.

Weixin Liang, Yuhui Zhang, Yongchan Kwon, Serena Yeung, and James Zou. Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. In *Advances in Neural Information Processing Systems*, volume 35, pages 17612–17625, 2022.

Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and Qiang Liu. Conflict-averse gradient descent for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 34, pages 18878–18890, 2021.

Massimiliano Mancini, Muhammad Ferjad Naeem, Yongqin Xian, and Zeynep Akata. Learning graph embeddings for open world compositional zero-shot learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):6565–6580, 2022.

Ishan Misra, Abhinav Gupta, and Martial Hebert. From red wine to red tomato: Composition with context. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 1792–1801, 2017.

Tushar Nagarajan and Kristen Grauman. Attributes as operators: Factorizing unseen attribute-object compositions. In *Proceedings of the European Conference on Computer Vision*, pages 169–185, 2018.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding, 2018.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021.

Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Mingxing Tan and Quoc V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning*, pages 6105–6114. PMLR, 2019.

Zirui Wang, Yulia Tsvetkov, Orhan Firat, and Yuan Cao. Gradient vaccine: Investigating and improving multi-task optimization in massively multilingual models. In *International Conference on Learning Representations*, 2021.

Linjie Xing and Yu Qiao. Deepwriter: A multi-stream deep cnn for text-independent writer identification. In *Proceedings of the 15th International Conference on Frontiers in Handwriting Recognition*, pages 584–589. IEEE, 2016.

Aron Yu and Kristen Grauman. Fine-grained visual comparisons with local learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 192–199, 2014.

Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 5824–5836, 2020.

A Appendix

We start our supplementary material with a simple 3-d toy example.

B A 3D Toy Example Illustrating Alignment Regimes

To build intuition, we construct a minimal three-dimensional example that mirrors the gradient structure used in the main paper. The goal is not to model a real dataset, but to show how changing the alignment between the visual prototype and the semantic target can change the sign of the gradient interaction.

B.1 Setup and Motivation

Consider an embedding space $\mathbb{R}^3$. We interpret the first coordinate as the primary visual direction that separates the target class, while the remaining coordinates represent competing or irrelevant semantic directions.

Let the visual prototype for the target class be

$$w_y = (1, 0, 0).$$

This means that, according to the primary classification objective, the target class lies along the first coordinate axis.

We first consider the strong-alignment case, where the semantic target agrees with the visual prototype:

$$t_y = (1, 0, 0).$$

Thus, the semantic anchor and the visual prototype point in the same direction.

We choose two non-target semantic anchors:

$$t_1 = (0, 1, 0), \qquad t_2 = (0, 0, 1).$$

These are orthogonal to the target direction and represent competing semantic concepts.

B.2 Softmax Weights

The auxiliary gradient has the form

$$g_2 \approx \sum_k q_k t_k - t_y.$$

Here, $\sum_k q_k t_k$ is the softmax-weighted average of semantic anchors predicted by the model, while t_y is the correct semantic anchor.

We set

$$q_y = 0.8, \qquad q_1 = 0.1, \qquad q_2 = 0.1.$$

This represents a confident but imperfect prediction: most probability mass is assigned to the correct semantic anchor, while a small amount remains on two competing anchors.

Therefore,

$$\sum_k q_k t_k = 0.8(1,0,0) + 0.1(0,1,0) + 0.1(0,0,1) = (0.8, 0.1, 0.1).$$

The auxiliary gradient becomes

$$g_2 = \sum_k q_k t_k - t_y = (0.8, 0.1, 0.1) - (1,0,0) = (-0.2, 0.1, 0.1).$$

This vector points mostly in the negative target direction, with small components toward competing semantic directions.

B.3 Primary Gradient

For simplicity, assume the primary classification gradient is dominated by attraction toward the target visual prototype:

$$g_1 = -w_y = (-1, 0, 0).$$

This is the direction that moves the representation toward the visual target prototype.

Thus, in the strong-alignment case, both g_1 and g_2 have negative first-coordinate components. They therefore point in broadly similar directions.

B.4 Case 1: Strong Alignment

We compute the gradient interaction:

$$\langle g_1, g_2 \rangle = (-1, 0, 0) \cdot (-0.2, 0.1, 0.1) = 0.2.$$

Since this value is positive,

$$\cos(g_1, g_2) > 0.$$

This corresponds to the cooperative regime. The primary objective and the auxiliary objective both encourage movement in compatible directions. Geometrically, this happens because the visual prototype w_y and the semantic target t_y are aligned.

B.5 Case 2: Weak Alignment

Weak alignment. Now consider a weak-alignment case where the semantic target is poorly aligned with the visual prototype:

$$t_y = (0.2, 0.8, 0).$$

To model semantic confusion, we choose a competing semantic anchor that is more aligned with the visual prototype direction:

$$t_1 = (0.8, 0.1, 0.1), \qquad t_2 = (0, 0, 1).$$

We assume the auxiliary prediction is dominated by the competing anchor:

$$q_y = 0, \qquad q_1 = 1, \qquad q_2 = 0.$$

Then

$$\sum_k q_k t_k = 0(0.2, 0.8, 0) + 1(0.8, 0.1, 0.1) + 0(0, 0, 1) = (0.8, 0.1, 0.1).$$

The auxiliary gradient becomes

$$g_2 = \sum_k q_k t_k - t_y = (0.8, 0.1, 0.1) - (0.2, 0.8, 0) = (0.6, -0.7, 0.1).$$

Using the same primary gradient

$$g_1 = (-1, 0, 0),$$

the interaction evaluates to

$$\langle g_1, g_2\rangle = (-1, 0, 0) \cdot (0.6, -0.7, 0.1) = -0.6 < 0.$$

Thus, the gradients conflict in the weak-alignment regime.

Hence,

$$\cos(g_1, g_2) < 0.$$

This is the conflict regime. The primary gradient wants to move the representation toward the visual prototype direction, while the auxiliary gradient pulls it away from that direction because the semantic target is misaligned.

B.6 Interpretation

The only essential change between the two cases is the alignment between w_y and t_y:

$$\langle w_y, t_y\rangle = \begin{cases} 1, & \text{strong alignment,} \\ 0.2, & \text{weak alignment.} \end{cases}$$

When w_y and t_y are aligned, the target-alignment term dominates and the gradients cooperate. When they are weakly aligned, the auxiliary gradient contains a component that opposes the primary gradient, causing negative interaction.

This toy example illustrates the mechanism studied in the main paper:

$$\text{semantic alignment} \quad \Longrightarrow \quad \text{sign and stability of gradient interaction.}$$

In writer identification, the auxiliary text space may not align well with the fine-grained visual directions required to distinguish writers, leading to weak or unstable interaction. In compositional image classification, semantic labels such as attribute–object pairs align more naturally with visual structure, leading to cooperative gradients.

C Sketch of Proof.

From the gradient decomposition, we write the gradients of the primary and auxiliary objectives with respect to the shared representation as

$$g_1 \approx \sum_j p_j w_j - w_y, \tag{2}$$

$$g_2 \approx \sum_k q_k t_k - t_y, \tag{3}$$

where w_y is the target class prototype and t_y is the target semantic anchor.

The gradient interaction is

$$\langle g_1, g_2\rangle = \left\langle \sum_j p_j w_j - w_y,\ \sum_k q_k t_k - t_y \right\rangle. \tag{4}$$

Using bilinearity of the inner product, we first expand with respect to the first argument:

$$\langle g_1, g_2\rangle = \left\langle \sum_j p_j w_j,\ \sum_k q_k t_k - t_y \right\rangle - \left\langle w_y,\ \sum_k q_k t_k - t_y \right\rangle. \tag{5}$$

Expanding each term again gives

$$\begin{aligned} \langle g_1, g_2\rangle = &\left\langle \sum_j p_j w_j,\ \sum_k q_k t_k \right\rangle - \left\langle \sum_j p_j w_j,\ t_y \right\rangle \\ &- \left\langle w_y,\ \sum_k q_k t_k \right\rangle + \langle w_y, t_y\rangle. \end{aligned} \tag{6}$$

Since scalars can be pulled outside inner products, we obtain

$$\left\langle \sum_j p_j w_j,\ \sum_k q_k t_k \right\rangle = \sum_{j,k} p_j q_k \langle w_j, t_k \rangle, \tag{7}$$

$$\left\langle \sum_j p_j w_j,\ t_y \right\rangle = \sum_j p_j \langle w_j, t_y \rangle, \tag{8}$$

$$\left\langle w_y,\ \sum_k q_k t_k \right\rangle = \sum_k q_k \langle w_y, t_k \rangle. \tag{9}$$

Substituting these terms back, we get

$$\langle g_1, g_2 \rangle = \langle w_y, t_y \rangle - \sum_j p_j \langle w_j, t_y \rangle - \sum_k q_k \langle w_y, t_k \rangle + \sum_{j,k} p_j q_k \langle w_j, t_k \rangle. \tag{10}$$

To make the target-alignment contribution explicit, we separate the target terms from the sums. Since the sums over j and k include $j = y$ and $k = y$, respectively, we write

$$\sum_j p_j \langle w_j, t_y \rangle = p_y \langle w_y, t_y \rangle + \sum_{j \neq y} p_j \langle w_j, t_y \rangle,$$

and

$$\sum_k q_k \langle w_y, t_k \rangle = q_y \langle w_y, t_y \rangle + \sum_{k \neq y} q_k \langle w_y, t_k \rangle.$$

Substituting these decompositions into Eq. (10) yields

$$\begin{aligned} \langle g_1, g_2 \rangle = (1 - p_y - q_y) \langle w_y, t_y \rangle &- \sum_{j \neq y} p_j \langle w_j, t_y \rangle - \sum_{k \neq y} q_k \langle w_y, t_k \rangle \\ &+ \sum_{j,k} p_j q_k \langle w_j, t_k \rangle. \end{aligned} \tag{11}$$

The final cross term also contains target-related components. Expanding it further,

$$\sum_{j,k} p_j q_k \langle w_j, t_k \rangle = p_y q_y \langle w_y, t_y \rangle + \sum_{j \neq y} p_j q_y \langle w_j, t_y \rangle + \sum_{k \neq y} p_y q_k \langle w_y, t_k \rangle + \sum_{j \neq y, k \neq y} p_j q_k \langle w_j, t_k \rangle.$$

Therefore, the coefficient of the target-alignment term becomes

$$1 - p_y - q_y + p_y q_y = (1 - p_y)(1 - q_y).$$

Thus, the interaction can be rewritten as

$$\begin{aligned} \langle g_1, g_2 \rangle = {} & (1 - p_y)(1 - q_y) \langle w_y, t_y \rangle \\ & - \sum_{j \neq y} p_j (1 - q_y) \langle w_j, t_y \rangle - \sum_{k \neq y} q_k (1 - p_y) \langle w_y, t_k \rangle \\ & + \sum_{j \neq y, k \neq y} p_j q_k \langle w_j, t_k \rangle. \end{aligned} \tag{12}$$

This decomposition separates the true target-alignment contribution from non-target and cross-class interactions. Since $(1 - p_y)(1 - q_y) \geq 0$, the target-alignment term contributes positively whenever the visual prototype w_y and semantic target t_y are aligned. The remaining terms collect interactions involving non-target prototypes or non-target semantic anchors. These terms may have mixed signs, but they act as competing contributions against the target-alignment term. This motivates approximating the expected gradient interaction by a dominant target-alignment component minus aggregate interference from non-target interactions.

Connection to Alignment gap. This motivates defining the alignment gap

$$\Delta = \mathbb{E}[\mathrm{sim}(z, t_y)] - \mathbb{E}[\mathrm{sim}(z, t_{\neg y})],$$

,where $t_{\neg y}$ denotes non-target semantic anchors. The alignment gap captures the separation between target and non-target similarities.

Thus, under the clustering approximation, the alignment gap provides an empirical proxy for the balance between target-aligned and non-target contributions to the expected gradient interaction, with larger Δ corresponding to more cooperative (positive) interaction and smaller Δ corresponding to increased interference from non-target terms.

Link between representation and prototypes. Using the clustering assumption $z \approx w_y$ introduced earlier, we relate prototype–anchor interactions to representation-level similarities. Thus, this alignment gap acts as an empirical proxy for the relative dominance of the target-alignment term over non-target contributions. Large Δ corresponds to cooperative gradients, while small Δ corresponds to unstable or conflicting gradient interaction.

D Derivation of the First-Order Gradient Interaction

The following text provides additional mathematical analysis to further interpret the alignment-controlled gradient interaction derived in Appendix C. These results complement the main derivation by examining how different components contribute to the observed regimes. Here, we provide the derivation of the first-order interaction used in the main paper.

Let the joint objective be

$$\mathcal{L}(\theta) = \mathcal{L}_1(\theta) + \lambda \mathcal{L}_2(\theta),$$

where $\lambda > 0$ controls the contribution of the auxiliary objective. Let

$$g_1 = \nabla_\theta \mathcal{L}_1(\theta), \qquad g_2 = \nabla_\theta \mathcal{L}_2(\theta).$$

For a small perturbation $\Delta\theta$, the first-order Taylor expansion of each loss is

$$\mathcal{L}_1(\theta + \Delta\theta) \approx \mathcal{L}_1(\theta) + g_1^\top \Delta\theta,$$

and

$$\mathcal{L}_2(\theta + \Delta\theta) \approx \mathcal{L}_2(\theta) + g_2^\top \Delta\theta.$$

Therefore, the combined objective satisfies

$$\begin{aligned}
\mathcal{L}(\theta + \Delta\theta) &= \mathcal{L}_1(\theta + \Delta\theta) + \lambda \mathcal{L}_2(\theta + \Delta\theta) \\
&\approx \mathcal{L}_1(\theta) + g_1^\top \Delta\theta + \lambda \left(\mathcal{L}_2(\theta) + g_2^\top \Delta\theta \right) \\
&= \mathcal{L}_1(\theta) + \lambda \mathcal{L}_2(\theta) + (g_1 + \lambda g_2)^\top \Delta\theta \\
&= \mathcal{L}(\theta) + (g_1 + \lambda g_2)^\top \Delta\theta.
\end{aligned}$$

Thus, the effective first-order update direction is

$$g = g_1 + \lambda g_2.$$

The interaction between the two objectives is determined by the inner product

$$\langle g_1, g_2 \rangle.$$

To see this, consider the squared norm of the combined gradient:

$$\begin{aligned}
\|g_1 + \lambda g_2\|^2 &= (g_1 + \lambda g_2)^\top (g_1 + \lambda g_2) \\
&= g_1^\top g_1 + 2\lambda g_1^\top g_2 + \lambda^2 g_2^\top g_2 \\
&= \|g_1\|^2 + 2\lambda \langle g_1, g_2 \rangle + \lambda^2 \|g_2\|^2.
\end{aligned}$$

Hence, when $\langle g_1, g_2\rangle > 0$, the objectives reinforce each other. When $\langle g_1, g_2\rangle < 0$, the objectives partially cancel. The critical interaction scale is obtained by minimizing $\|g_1 + \lambda g_2\|^2$ with respect to λ:

$$\frac{\partial}{\partial\lambda}\|g_1 + \lambda g_2\|^2 = 2\langle g_1, g_2\rangle + 2\lambda\|g_2\|^2.$$

Setting this derivative to zero gives

$$\lambda^\star = -\frac{\langle g_1, g_2\rangle}{\|g_2\|^2}.$$

This quantity characterizes the scale at which the auxiliary gradient can maximally cancel the primary gradient when the two gradients are in conflict. These results further support the interpretation that the balance between target and non-target contributions, captured by the alignment gap Δ, governs the sign and stability of gradient interaction.

E Cross-Domain Study: Compositional Image Classification

E.1 Setup

We evaluate the proposed alignment-based diagnostic (as used for writer identification problem) on UT-Zappos50K compositional image classification [Yu and Grauman, 2014] using CLIP-based text supervision [Radford et al., 2021]. We report results for ViT-B/16 [Dosovitskiy et al., 2021] and EfficientNet-B0 [Tan and Le, 2019]. Here, each class corresponds to an attribute–object composition (a, o), and the class-level semantic anchor is obtained from CLIP text embeddings of the corresponding composition. The dataset is parsed from the official split files and metadata, and image paths are resolved from the `images/_images` directory. We use the train split for optimization and the validation split for the final target/non-target similarity and subspace-energy diagnostics.

Semantic anchors:- We use the frozen CLIP ViT-B/32 text encoder [Radford et al., 2021] to construct one semantic anchor per attribute–object composition. For a class (a, o), we encode four prompt templates,

$$\text{“a photo of a } a\, o\text{”}, \quad \text{“this is a } a\, o\text{”}, \tag{10}$$

$$\text{“a } a \text{ colored } o\text{”}, \quad \text{“an image showing } a\, o\text{”}. \tag{11}$$

The four CLIP text embeddings are individually ℓ_2-normalized, averaged, and normalized again to obtain the class anchor $t_c \in \mathbb{R}^{512}$. Stacking all anchors gives $T = [t_1, \ldots, t_C] \in \mathbb{R}^{512\times C}$.

Visual model:- We report results for two ImageNet-pretrained visual backbones: ViT-B/16 [Dosovitskiy et al., 2021] and EfficientNet-B0 [Tan and Le, 2019]. For both models, the original classification head is removed and replaced by a two-layer projection head that maps the visual feature to a 512-dimensional normalized representation z. A bias-free linear classifier is then applied to z for the primary compositional classification objective.

Objective and optimization:- For an image x_i with composition label y_i, the primary objective is cross-entropy over the learned classifier logits, while the auxiliary objective is an InfoNCE-style alignment loss against the frozen CLIP text-anchor matrix:

$$\mathcal{L}_{\text{nce}} = -\log\frac{\exp(z_i^\top t_{y_i}/\tau)}{\sum_{c=1}^{C}\exp(z_i^\top t_c/\tau)}, \qquad \tau = 0.07. \tag{12}$$

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{nce}}, \tag{13}$$

with $\lambda \in \{0, 0.01, 0.05, 0.10, 0.20, 0.50\}$. We train each model for 30 epochs using Adam with learning rate 10^{-4} and batch size 64. Images are resized to 224×224; training uses random horizontal flips and color jitter, while validation uses deterministic resizing. All images are normalized with ImageNet mean and standard deviation.

Table 1: Qualitative summary of the predictive relationship between alignment gap and gradient interaction. Exact values depend on the backbone, seed, and auxiliary weight, but the regimes are stable across the reported diagnostics.

Task / Backbone	Alignment gap	Mean cosine	Negative fraction	Regime
Writer / FragNet	small / near-zero	negative-shifted	high	Conflict
Writer / Transformer	small / near-zero	near zero	moderate	Neutral
Writer / DeepWriter	small positive	mildly positive	lower	Partial cooperation
UT-Zappos / ViT-B/16	large (≈ 0.17)	positive	negligible	Cooperation
UT-Zappos / EfficientNet-B0	large (≈ 0.17)	positive	negligible	Cooperation

Diagnostics:- At each training epoch, we compute the parameter-space cosine similarity between the classification gradient and the text-alignment gradient before the update. We also record the negative-gradient fraction, training accuracy, validation accuracy, target and non-target text similarities, and visual/text subspace energies. The target/non-target similarity and subspace-energy distributions shown in the final diagnostic plots are computed on the validation split. We use the accuracy curves as supporting diagnostics for optimization stability, while the main evidence for cooperation comes from the gradient-cosine distributions, negative-fraction, target/non-target similarity, and subspace-energy analyses.

Reported backbones:- All compositional-classification plots and claims in this version correspond to the two reported backbones, namely ViT-B/16 [Dosovitskiy et al., 2021] and EfficientNet-B0 [Tan and Le, 2019].

E.2 Effect of λ on Accuracy

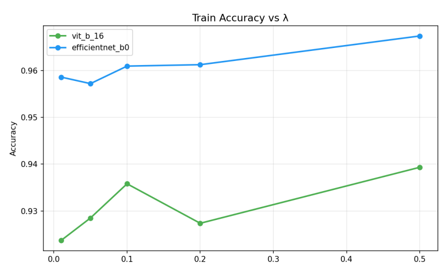


Figure 7: Training accuracy vs λ in compositional image classification. Performance remains stable in the cooperative regime.

As shown in Fig. 7, the training accuracy remains stable across λ for ViT-B/16 and EfficientNet-B0, suggesting that strong alignment prevents the auxiliary objective from destabilizing optimization. We therefore interpret the UT-Zappos compositional classification experiment as a strong-alignment regime: semantic supervision follows the visual decision structure, gradient interaction is consistently cooperative, and optimization remains robust over a broad range of auxiliary weights. We do not use this training-accuracy plot alone as evidence of improved generalization; rather, it serves as an optimization diagnostic complementary to the gradient-cosine, negative-fraction, target/non-target similarity, and subspace-energy analyses.

E.3 Alignment Gap as a Predictive Diagnostic

Table 1 summarizes the qualitative relationship observed across the two empirical regimes. Writer identification has small target–non-target separation, and its gradient interaction depends strongly on the visual backbone. UT-Zappos compositional classification has a much larger alignment gap and consistently cooperative gradient interaction for the two reported backbones. This supports the central claim that Δ acts as a task-supervision compatibility diagnostic: small Δ predicts unstable or conflicting auxiliary interaction, whereas large Δ predicts cooperative interaction.

E.4 Subspace Energy Analysis

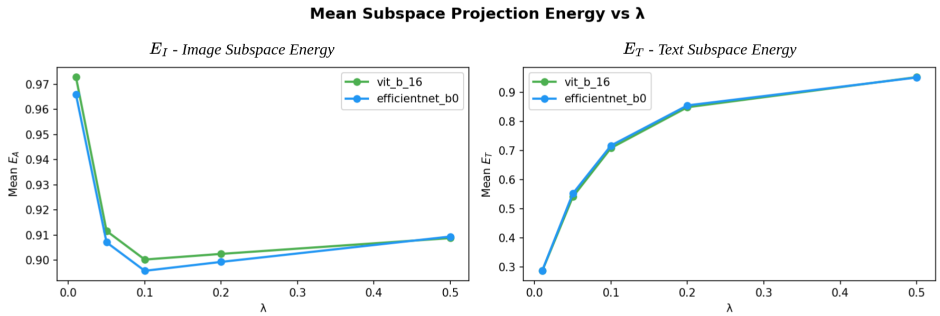


Figure 8: Subspace energy vs λ.

Figure 8 plots projection energies onto the visual subspace (E_I) and semantic subspace (E_T) as λ increases. As λ grows, E_T increases rapidly before plateauing, while E_I decreases slightly and then stabilizes. This indicates a redistribution of representational energy toward the semantic subspace as the auxiliary objective is emphasized. Importantly, both curves saturate rather than diverge, demonstrating that the representation geometry remains stable. This stability of representation geometry under varying λ is consistent with the strong-alignment regime, where the alignment gap Δ remains large and gradient interaction remains cooperative.

F Additional Experimental Results

This appendix provides additional analysis to further interpret the alignment-controlled gradient interaction. While Appendix C and D focus on the primary derivation, we include these results to offer complementary perspectives on how alignment influences optimization behavior.

F.1 Full Writer Identification Diagnostics

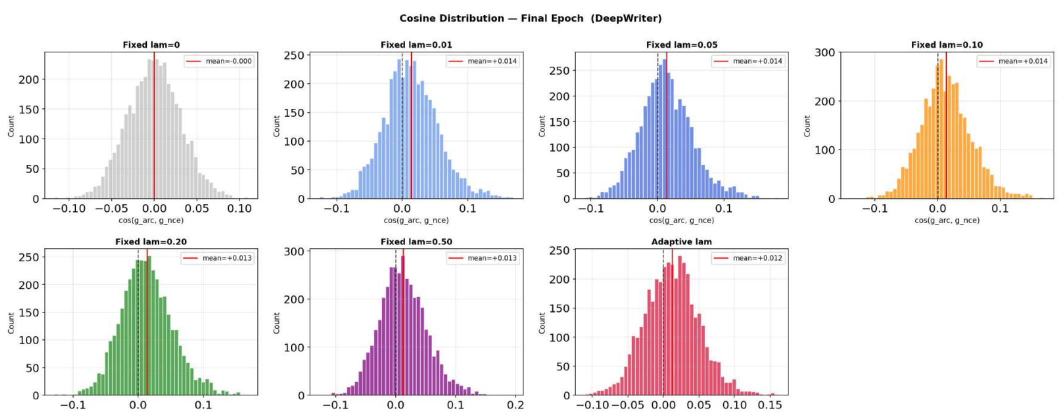


Figure 9: Gradient cosine similarity distributions across fixed λ values $(0, 0.01, 0.05, 0.10, 0.20, 0.50)$ and an adaptive λ for the DeepWriter backbone.

Figure 8 illustrates that distributions are consistently centered slightly above zero (mean ≈ 0.011–0.014), indicating weak but positive gradient interaction. As λ increases, the distributions shift modestly toward the positive region without increasing variance significantly, suggesting improved cooperation without instability. The adaptive λ further stabilizes the distribution. Overall, DeepWriter exhibits partial alignment, representing the most cooperative regime among the writer identification backbones under weak semantic alignment. This corresponds to a slightly positive alignment gap Δ, explaining the partial cooperation observed in this backbone.

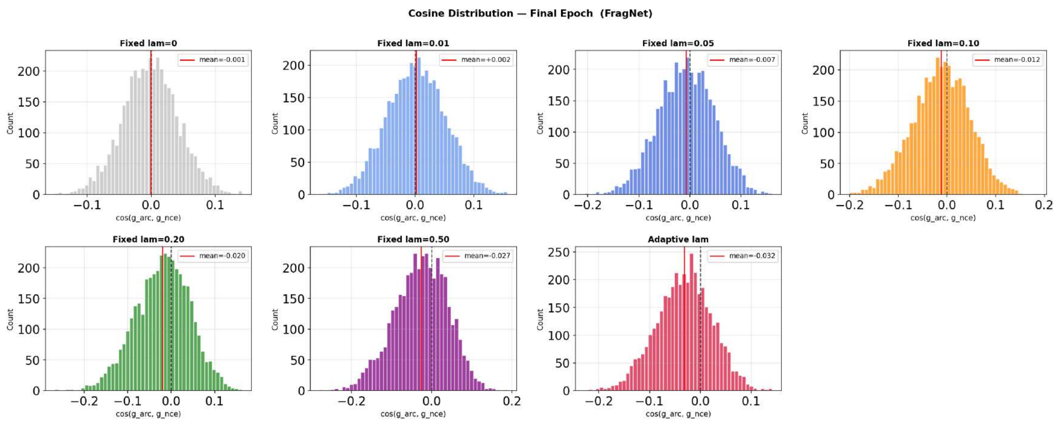


Figure 10: Gradient cosine similarity distributions across fixed λ values $(0, 0.01, 0.05, 0.10, 0.20, 0.50)$ and an adaptive λ for the Fragnet backbone.

In Figure 9, the distributions are centered near or below zero, with increasingly negative means at higher λ (e.g., ≈ -0.020 at $\lambda = 0.20$ and ≈ -0.027 at $\lambda = 0.50$), indicating a conflict-dominated regime. The negative shift reflects a near-zero or negative effective alignment gap, where non-target contributions dominate. As λ increases, the distributions shift leftward, demonstrating that stronger auxiliary weighting exacerbates gradient conflict. While adaptive λ reduces extreme variance, it does not recover positive alignment. These results confirm that FragNet is highly sensitive to loss weighting under weak alignment.

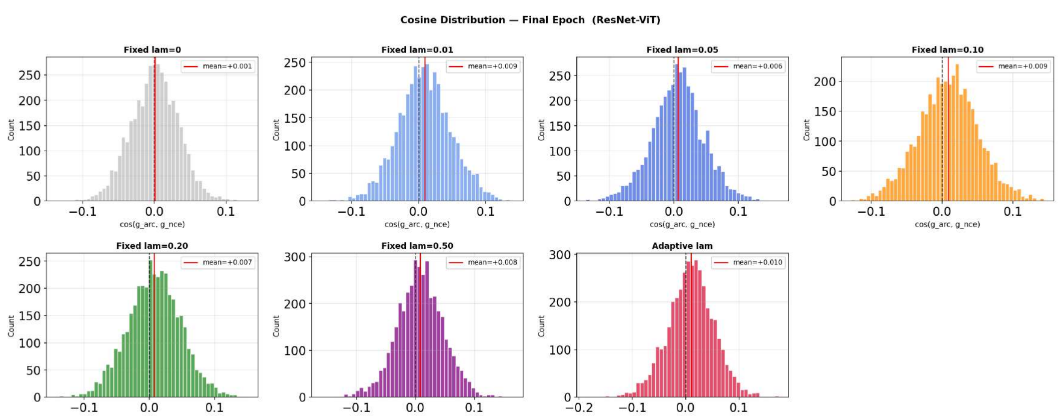


Figure 11: Gradient cosine similarity distributions across fixed λ values $(0, 0.01, 0.05, 0.10, 0.20, 0.50)$ and an adaptive λ for the ResNet-ViT backbone.

Figure 10 shows gradient cosine similarity distributions for the Transformer backbone. The distributions remain tightly centered around zero (mean ≈ 0.001–0.007) across all λ values. The near-zero mean is consistent with $\Delta \approx 0$, indicating a neutrality regime where neither target nor non-target contributions dominate. Unlike DeepWriter (slightly positive) and FragNet (negative), the Transformer consistently exhibits stable but non-cooperative behavior. The adaptive λ slightly reduces variance but does not shift the mean, reinforcing that the model operates in a neutrality regime under weak alignment.

F.2 Post-hoc Analysis of Alignment Gap

To further examine the proposed alignment-controlled interaction view, we perform a quantitative post-hoc analysis of the relationship between the alignment gap Δ and gradient interaction statistics under different auxiliary-loss weighting settings.

The statistics reported here complement the distributional analysis presented in Section 4.2. While MeanCos summarizes the average gradient interaction, the cosine distributions in the main paper reveal the variability and stability of interaction across samples. As a result, positive average cosine interaction does not necessarily imply uniformly cooperative optimization behavior, particularly in weak-alignment regimes where interaction distributions remain strongly architecture-dependent.

In Tables 2–4, the sample-level alignment gap statistics (Δ mean/std), mean gradient cosine similarity (MeanCos), fraction of samples with negative cosine interaction (NegFrac%), and best validation accuracy (BestVal%) for the DeepWriter, FragNet, and ResNet-ViT backbones, have been reported respectively.

For the DeepWriter backbone, MeanCos remains close to zero and the fraction of negative interactions remains relatively high across different auxiliary-loss settings, consistent with the weak-alignment regime discussed in the main paper. We can see that FragNet exhibits highly unstable and asymmetric interaction behavior. Although the average cosine interaction becomes mildly positive under auxiliary supervision, the interaction distributions remain strongly left-skewed with substantial negative-gradient mass, indicating frequent conflict at the sample level.

Interestingly, the ResNet-ViT backbone exhibits consistently stronger positive cosine interaction and lower fractions of negative interaction.

Overall, the post-hoc analysis supports the broader interpretation that representation-level alignment strongly influences gradient interaction behavior, while the resulting optimization dynamics remain architecture-dependent in weak-alignment regimes.

Table 2: Post-hoc analysis results for the DeepWriter backbone. The table reports the sample-level alignment gap (Δ mean, std), gradient cosine similarity (MeanCos), the fraction of samples with negative gradient cosine (NegFrac%), and best validation accuracy (BestVal%) under different fixed and adaptive λ settings on the full validation set with unit-normalized embeddings.

Setup	**Δ mean**	**Δ std**	**MeanCos**	**NegFrac%**	**BestVal%**
ArcFace only	+0.0001	0.0001	-0.0005	50.2%	86.32%
Fixed λ=0	+0.0001	0.0001	+0.0017	48.3%	86.04%
Fixed λ=0.01	+0.0002	0.0001	+0.0112	43.8%	85.75%
Fixed λ=0.05	+0.0002	0.0002	+0.0127	41.5%	85.61%
Fixed λ=0.10	+0.0002	0.0002	+0.0134	40.9%	86.47%
Fixed λ=0.20	+0.0003	0.0002	+0.0141	39.6%	86.61%
Fixed λ=0.50	+0.0002	0.0002	+0.0118	42.7%	85.90%

Table 3: Post-hoc analysis results for the FragNet backbone. The table reports the sample-level alignment gap (Δ mean, std), gradient cosine similarity (MeanCos), the fraction of samples with negative gradient cosine (NegFrac%), and best validation accuracy (BestVal%) under different fixed and adaptive λ settings on the full validation set with unit-normalized embeddings.

Setup	**Δ mean**	**Δ std**	**MeanCos**	**NegFrac%**	**BestVal%**
ArcFace only	+0.0001	0.0001	-0.0008	50.9%	90.03%
Fixed λ=0	+0.0002	0.0002	+0.0017	48.3%	90.03%
Fixed λ=0.01	+0.0002	0.0000	+0.0248	34.2%	90.17%
Fixed λ=0.05	+0.0003	0.0002	+0.0239	33.3%	90.60%
Fixed λ=0.10	+0.0001	0.0002	+0.0231	33.6%	89.74%
Fixed λ=0.20	+0.0002	0.0002	+0.0246	32.8%	89.60%
Fixed λ=0.50	+0.0002	0.0001	+0.0205	36.2%	89.89%
Adaptive λ	+0.0002	0.0002	+0.0170	37.3%	90.31%

Table 4: Post-hoc analysis results for the ResNet-ViT backbone. The table reports the sample-level alignment gap (Δ mean, std), gradient cosine similarity (MeanCos), the fraction of samples with negative gradient cosine (NegFrac%), and best validation accuracy (BestVal%) under different fixed and adaptive λ settings on the full validation set with unit-normalized embeddings.

Setup	**Δ mean**	**Δ std**	**MeanCos**	**NegFrac%**	**BestVal%**
ArcFace only	-0.0000	0.0001	-0.0011	51.6%	91.60%
Fixed λ=0	-0.0001	0.0001	-0.0011	50.4%	92.02%
Fixed λ=0.01	+0.0001	0.0004	+0.0189	36.3%	91.60%
Fixed λ=0.05	+0.0000	0.0004	+0.0200	35.2%	91.88%
Fixed λ=0.10	-0.0001	0.0002	+0.0188	36.2%	91.88%
Fixed λ=0.20	-0.0000	0.0002	+0.0218	34.3%	91.88%
Fixed λ=0.50	-0.0001	0.0002	+0.0197	34.8%	92.02%
Adaptive λ	+0.0001	0.0002	+0.0211	34.0%	92.59%

F.3 Additional Gradient Cosine Analysis

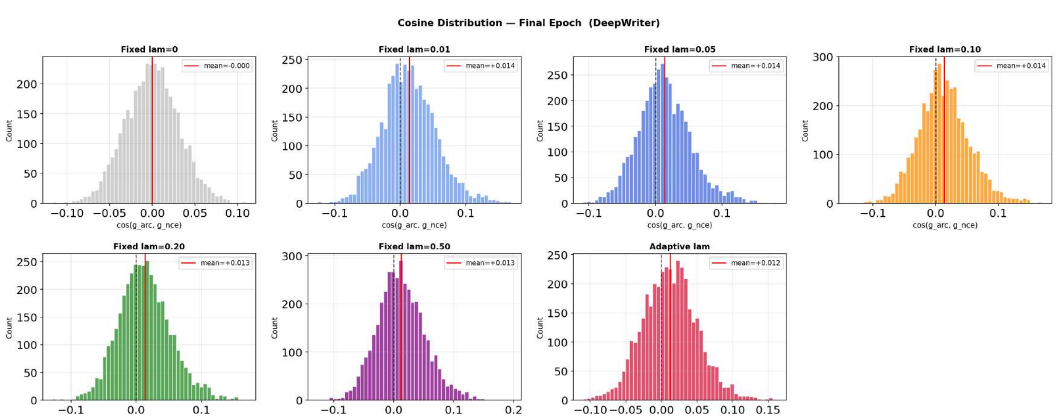


Figure 12: Additional cosine analysis: DeepWriter.

These additional results further validate that gradient interaction behavior is consistently governed by the alignment gap Δ across architectures and hyperparameter settings. Figure 12 provides additional gradient cosine distributions for DeepWriter under extended settings. The distributions remain consistently shifted toward positive values (mean ≈ 0.011–0.014), with minimal sensitivity to λ. The persistent positive shift indicates a stable positive Δ, corresponding to partial alignment. The adaptive λ again yields the most concentrated distribution. These results confirm that the observed partial alignment in DeepWriter is robust and not dependent on specific experimental conditions.

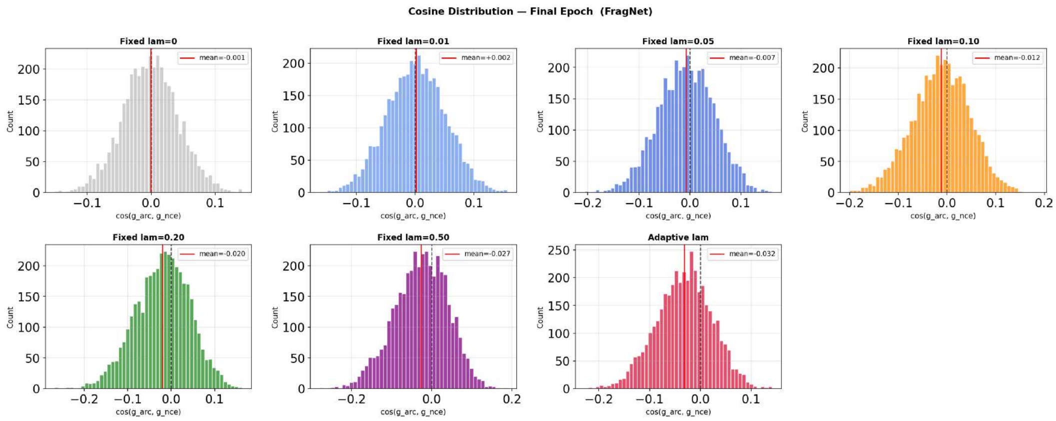


Figure 13: Additional cosine analysis: FragNet.

Figure 13 extends the cosine analysis for FragNet. The distributions remain centered near or below zero, with increasingly negative shifts at higher λ values. The persistent leftward drift confirms strong gradient conflict, and the variability across settings highlights instability. These results reinforce that FragNet consistently operates in a conflict-dominated regime under weak alignment.

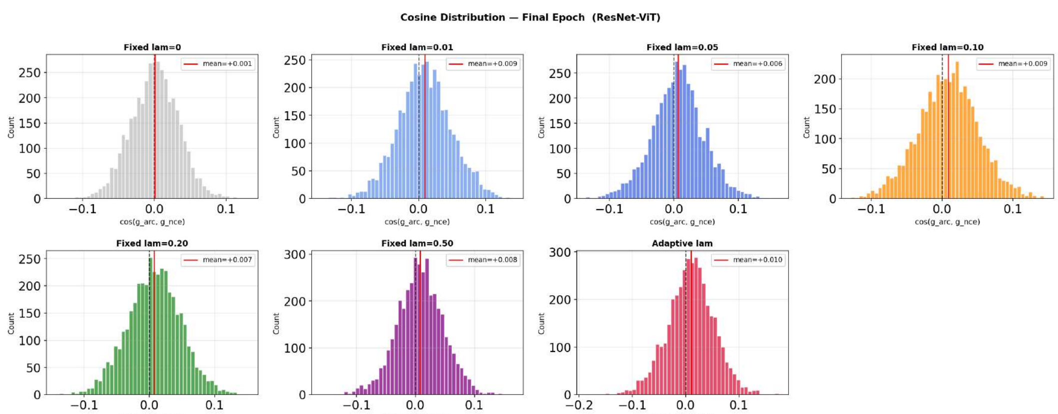


Figure 14: Additional cosine analysis: Transformer.

Figure 14 shows extended cosine similarity distributions for the Transformer. The distributions remain tightly centered around zero across all λ values, with no systematic drift. This confirms that the near-neutral interaction observed earlier is stable and reproducible. The Transformer therefore occupies a consistent neutrality regime, distinct from both cooperative (DeepWriter) and conflicting (FragNet) behaviors.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

Justification: Sufficient experimental results support the claim.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification:Limitations and future work are mentioned in the paper.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: All assumptions used in the first-order interaction analysis are explicitly stated in Section 3 and Appendix C. Derivations and proof sketches are provided there.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.

- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Github link with the code will be published along with the data splits.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: Data and Code will be available in project git repository in the future once the paper gets accepted.

   Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Clearly mentioned data splits and seeds.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: We report statistical variability through sample-level summary statistics of the proposed alignment gap Δ, including mean and standard deviation under different optimization settings. In addition, we report aggregate gradient interaction statistics such as mean cosine similarity and the fraction of samples with negative cosine interaction. These statistics support the empirical trends discussed in the paper.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.

- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: Detailed information about compute resources, training configurations, and runtime requirements will be included in the public code repository upon release to facilitate reproducibility of the experiments.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: Mentioned usage of LLM in the paper.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: This a core technical study.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: Used publicly available datasets.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: Datasets have been cited.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: No new assets are released.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: No human subjects were involved.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects were involved.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: Already mentioned in paper acknowledgement section.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.